

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi – 590018



Major Project Report On

“Ghost Writer Ai: AI-Generated Text Detection Using Hybrid NLP and Gen ai”

Submitted in partial fulfillment of the requirements as a part of the curriculum,

Bachelors of Engineering in Artificial Intelligence and Machine Learning

Submitted by

Vikram S [1CR23AI411]
Gowtham M [1CR23AI401]
P Sai Kiran [1CR23AI078]
Parthib Dey [1CR23AI034]

Under the Guidance of

Ms. Sandhya Kumari

Assistant Professor

Department of Artificial Intelligence and Machine Learning



Department of Artificial Intelligence and Machine Learning

CMR INSTITUTE OF TECHNOLOGY

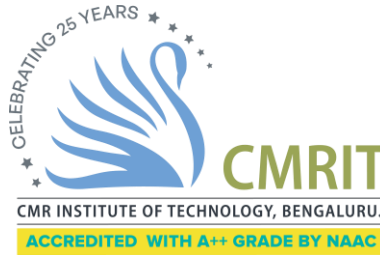
132, AECS Layout, Kundalahalli, ITPL Main Rd, Bengaluru – 560037

2025-2026

CMR INSTITUTE OF TECHNOLOGY

132, AECS Layout, Kundalahalli colony, ITPL Main Rd, Bengaluru-560037

Department of Artificial Intelligence and Machine Learning



CERTIFICATE

Certified that the Entrepreneurial Project entitled "**Ghost Writer Ai: AI-Generated Text Detection Using Hybrid NLP and Gen ai**" is presented by Vedanta Verma and Tanay Chaturvedi, bonafide students of CMR Institute of Technology in partial fulfillment of the requirements as a part of the curriculum, [Bachelors of Engineering in Artificial Intelligence and Machine Learning](#), of Visvesvaraya Technological University, Belagavi during the year **2025-26**.

It is certified that all correction/suggestion indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Entrepreneurial report has been approved as it satisfies the academic requirements in respect of the Technical Seminar prescribed for the bachelor of engineering degree.

Ms. Sandhya Kumari

Signature of the Guide

Dr. Shyam P Joy

Signature of the HOD

**Ghost Writer Ai: AI-
Generated Text Detection
Using Hybrid NLP and Gen
ai**

ABSTRACT

The project “Ghost Writer Ai:AI-Generated Text Detection Using Hybrid NLP and Gen Ai” introduces a hybrid framework designed to distinguish AI-generated content from human-written text using Natural Language Processing techniques and transformer-based deep learning architecture

The proposed framework combines stylometric analysis, TF-IDF vectorization, readability metrics, and contextual semantic embeddings generated using gen Ai. The system integrates traditional linguistic feature extraction with contextual semantic understanding to improve robustness, scalability, and detection accuracy.

The framework utilizes benchmark datasets including HC3 and DAIGT v2 under topic-based dataset splitting conditions to improve generalization capability and reduce information leakage. Experimental evaluation demonstrates that transformer-based models significantly outperform traditional machine learning approaches in contextual understanding, classification performance, and robustness against paraphrasing attacks.

The proposed system aims to provide a scalable and reliable AI-generated text detection framework for academic integrity verification, misinformation detection, plagiarism prevention, and digital content authentication systems.

ACKNOWLEDGEMENT

It is my proud privilege and duty to acknowledge the kind of help and guidance received from several people in preparation of this report. Apart from my own, the success of this report depends largely on the encouragement and guidelines of many others. It would have not been possible to prepare this report in this form without their valuable help, co-operation and guidance.

I would like to express my deep sense of gratitude to our Principal **Dr. Sanjay Jain**, CMR Institute of Technology College, Bangalore for his motivation and for creating an inspiring atmosphere in college by providing state of the art facilities for preparation and delivery of this report.

My sincere thanks to **Dr. Shyam P Joy**, Head of Department of Artificial Intelligence and Machine Learning, for his whole-hearted support in completion of the mini project.

I am highly indebted to my seminar guide Assistant Prof. **Sandhya Kumari**, for guiding and giving timely advices and suggestions in the successful completion of this seminar.

My sincere thanks to Assistant Prof. **Ms. B Saranya**, Entrepreneurial project coordinator for having supported the work related to this seminar.

Last but not least, I would like to put forward my heartfelt acknowledgement to all my classmates, friends and all those who have directly or indirectly provided their overwhelming support during my seminar work and the development of this report.

Vikram S [1CR23AI411]

Gowtham M [1CR23AI401]

P Sai Kiran [1CR23AI078]

Parthib Dey [1CR23AI079]

CONTENTS

Certificate	II
Title Page	III
Abstract	IV
Acknowledgement	V
CHAPTER 1. Introduction	
1.1 Background	1
1.2 Objective	1
CHAPTER 2. Review of Related Literature	
2.1 Literature Review	3
2.2 Problem Statement	3
2.3 Implementation	4
CHAPTER 3. Design and Implementation	
3.1 System Architecture	5
3.2 Working Principle	5
3.3 Dataset Collection	6
CHAPTER 4. Result and Analysis	
4.1 Experimental Evaluation	7
4.2 Comparative Analysis	7
4.3 System Behavior	8
CHAPTER 5. Conclusions & future enhancements	9

GROUP MEMBER LIST

SL NO	NAME OF STUDENT	USN
01	Vikram S	1CR23AI411
02	Gowtham M	1CR23AI401
03	P Sai Kiran	1CR23AI078
04	Parthib Dey	1CR23AI079

CHAPTER 1

INTRODUCTION

1.1 Background

Large Language Models are rapidly becoming essential tools in modern Artificial Intelligence applications. These systems are widely used in academic writing, research assistance, customer support, software development, journalism, and enterprise automation. Despite their advanced language generation capabilities, these models continue to suffer from hallucination, where generated outputs appear factually correct but are actually fabricated or unsupported.

Traditional plagiarism detection systems mainly rely on textual similarity matching and are ineffective against modern AI-generated content because LLMs generate contextually rich and unique responses instead of directly copying existing information.

AI-generated text detection has therefore emerged as a major research area in Natural Language Processing and Machine Learning. The proposed framework addresses this challenge through a hybrid architecture combining traditional NLP feature extraction and transformer-based contextual semantic analysis.

1.2 Objective

The primary objective of this project is to detect AI-generated text using a hybrid NLP and transformer-based framework.

The objectives include:

- Designing a hybrid AI-generated text detection framework
- Integrating stylometric analysis and TF-IDF vectorization
- Implementing transformer-based semantic classification
- Improving contextual understanding and robustness
- Evaluating benchmark datasets such as HC3 and DAIGT v2
- Reducing false classifications and improving scalability.

1.3 Motivation

The increasing adoption of generative AI systems has created serious concerns regarding misinformation, plagiarism, fake news generation, and academic dishonesty. Existing AI systems often lack reliable mechanisms to verify whether content is human-written or machine generated

detection framework capable of improving trustworthiness, transparency, and authenticity verification in digital systems.

CHAPTER 2

REVIEW OF RELATED LITERATURE

2.1 Literature Review

Several approaches have been proposed for AI-generated text detection. Traditional machine learning techniques such as Logistic Regression, Support Vector Machines, Random Forest, and XGBoost primarily rely on statistical and stylometric features including word frequency, readability scores, punctuation usage, and vocabulary richness.

Transformer-based architectures such as BERT, DistilBERT, and RoBERTa significantly improved contextual semantic understanding through self-attention mechanisms and contextual embeddings.

Recent hybrid frameworks combine traditional NLP feature extraction with transformer-based semantic analysis to improve robustness and scalability.

2.2 Problem Statement

Current AI systems generate highly convincing but factually unsupported content because they rely on probabilistic text generation instead of factual validation. Existing plagiarism detection systems and traditional NLP models fail to effectively distinguish AI-generated content from human-written text.

Therefore, there exists a strong need for a scalable hybrid AI-generated text detection framework capable of contextual semantic analysis, stylometric analysis, and robust classification across multiple domains and datasets.

2.3 Existing Challenges

AI-generated text detection presents multiple technical challenges:

Poor generalization across unseen topics

Continuous evolution of Large Language Models

Dataset quality and scalability limitations

Modern transformer-based language models increasingly mimic human writing patterns, making detection significantly more difficult.

2.4 Technical Papers by Each Member:

1. Detection of AI-Generated Text Using Linguistic Features and Machine Learning for Preserving Academic Integrity

Paper Link: <https://vectmag.com/sgsi/paper/view/540>

By Vikram S

Detection of AI-Generated Text Using Linguistic Features and Machine Learning for Preserving Academic Integrity

This paper focuses on detecting AI-generated text using linguistic and stylometric features combined with machine learning techniques to preserve academic integrity. The study analyzes characteristics such as vocabulary richness, readability, sentence structure, punctuation usage, and lexical diversity to distinguish human-written content from AI-generated text. ([arXiv][1])

The research highlights that machine learning models such as Random Forest and XGBoost achieved strong classification performance when trained on linguistic features extracted from textual data. The study also emphasizes the importance of Explainable AI (XAI) techniques for improving transparency and understanding the reasoning behind AI-generated text classification. The proposed framework aims to support plagiarism prevention, ethical AI usage, and academic content "Detecting AI-Generated Text in Educational Content: Leveraging Machine Learning and Explainable AI for Academic Integrity"

2. AI Generated Text Detection: <https://arxiv.org/pdf/2601.03812>

By Goutam M

ChatEval is a multi-agent evaluation framework that improves the reliability of AI-generated response evaluation using collaborative debate among multiple AI agents. Instead of depending on a single evaluator model, the framework simulates a human-like review process where different agents discuss, critique, and evaluate generated responses before arriving at a final decision. The system introduces role-based discussion and debate mechanisms to

improve alignment with human judgment in Natural Language Generation and reasoning tasks.

The paper demonstrates that collaborative multi-agent debate significantly improves consistency, fairness, and reasoning quality compared to traditional single-model evaluators. By exposing disagreement and critique among agents, the framework reduces evaluation bias and improves transparency. ChatEval also highlights the importance of debate-driven reasoning in improving AI alignment and trustworthy evaluation systems. However, the framework mainly focuses on evaluation tasks rather than real-time fact verification and still faces scalability and computational efficiency challenges in practical deployment.

3. An Empirical Study of AI Generated Text Detection Tools

Paper link: <https://arxiv.org/abs/2310.01423>

By Sai Kiran

This paper presents an empirical analysis of various AI-generated text detection tools used to identify content generated by Large Language Models such as ChatGPT. The study evaluates six popular AI detection systems including GPTZero, GPTKit, Originality.ai, Sapling, Writer, and Zylalab across multiple domains such as news articles, stories, abstracts, and product reviews.

The research highlights that detection accuracy varies significantly between tools, with performance ranging from approximately 55% to 97%. Among all evaluated systems, Originality.ai achieved the highest overall detection accuracy. The paper also emphasizes that AI-generated text detection remains challenging due to the evolving quality and contextual sophistication of modern Large Language Models.

4. Generative AI-Based Text Generation Methods Using Pre-Trained GPT-2 Model

Paper link: <https://arxiv.org/abs/2404.01786>

By Parthib Dey

This paper explores multiple Generative AI text generation techniques using the GPT-2 transformer model. The study analyzes decoding methods such as Greedy Search, Beam Search, Top-K Sampling, Top-P Sampling, Contrastive Search, and Locally Typical Sampling for generating contextually relevant and coherent text.

The research highlights how transformer-based language models learn linguistic patterns, grammar, and semantic relationships from large textual datasets to generate human-like content. The paper also compares the strengths and limitations of different text generation strategies based on coherence, diversity, and contextual relevance.

CHAPTER 3

DESIGN AND IMPLEMENTATION

3.1 System Architecture

The proposed framework follows a hybrid architecture designed for contextual semantic analysis and stylometric feature extraction.

1. Frontend Layer:

- Developed using React.js, HTML, CSS, and JavaScript.
- Provides text input and prediction visualization.
- Displays classification results and confidence scores.

2. Backend Layer:

- Implemented using Python and FastAPI.
- Handles preprocessing, feature extraction, and inference.
- Stores prediction logs and dataset processing workflows

3. AI Reasoning Layer:

- Contains NLP preprocessing modules.
- Integrates TF-IDF vectorization and stylometric analysis.
- Utilizes DistilBERT and RoBERTa for contextual embeddings.

3.2 Working Principle

The framework operates using the following workflow:

1. User submits textual input.
2. Text undergoes preprocessing.
3. Linguistic and stylometric features are extracted.
4. TF-IDF vectors are generated.
5. Transformer embeddings are created using DistilBERT or RoBERTa.
6. Classification models predict whether the content is human-written or AI-generated.
7. Final prediction is displayed to the user.

3.3 Dataset Collection

The proposed framework utilizes benchmark datasets including HC3 and DAIGT v2. These datasets contain both human-written and AI-generated textual samples across multiple domains and writing styles.

Topic-based dataset splitting is used to reduce information leakage and improve real-world evaluation performance.

3.4 Feature Extraction

The framework extracts multiple linguistic and statistical features including:

- TF-IDF vectors
- Sentence length
- Vocabulary richness
- Readability scores
- Lexical diversity
- Part-of-speech patterns
- Punctuation frequency

Transformer embeddings generated by DistilBERT and RoBERTa capture contextual semantic relationships within textual data.

3.5 Generative AI Integration

The proposed framework incorporates Generative AI concepts to improve AI-generated text detection and contextual semantic understanding. Large Language Models such as ChatGPT, Gemini, Claude, and LLaMA are analyzed as primary sources of machine-generated textual data.

The system studies linguistic behavior, semantic flow, response consistency, and writing patterns commonly associated with Generative AI systems. Transformer-based embeddings and contextual semantic analysis are utilized to identify repetitive sentence structures, probabilistic phrasing, stylometric inconsistencies, and unnatural coherence patterns typically observed in AI-generated content.

Generative AI techniques are also utilized during dataset preparation and adversarial testing. AI-generated samples are produced using multiple prompting strategies, paraphrasing techniques, and prompt engineering methods to improve the robustness and generalization capability of the detection framework against evolving Large Language Models.

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Experimental Evaluation

The proposed AI-generated text detection framework was evaluated using benchmark datasets under topic-based dataset splitting conditions to measure contextual understanding, semantic analysis capability, and robustness against AI-generated content. The experimental analysis demonstrated that transformer-based architectures significantly outperformed traditional machine learning approaches in detecting AI-generated text across multiple domains and writing styles. The integration of stylometric analysis, TF-IDF vectorization, and transformer embeddings improved classification consistency and contextual semantic understanding.

DistilBERT achieved strong performance while maintaining computational efficiency due to its lightweight architecture. RoBERTa demonstrated superior semantic representation and contextual learning capability, resulting in improved robustness against paraphrased and adversarial AI-generated content.

The hybrid framework successfully improved generalization capability across unseen topics and reduced information leakage through topic-based dataset splitting. Experimental results confirmed that combining traditional NLP techniques with transformer-based semantic modeling provides a scalable and effective solution for AI-generated text detection.

4.2 Comparative Analysis

The framework analyzed AI-generated text produced by models such as ChatGPT, Gemini, Claude, and LLaMA to identify linguistic and semantic patterns commonly associated with Generative AI systems.

Experimental analysis showed that AI-generated content often contains repetitive sentence structures, consistent grammatical patterns, and predictable semantic flow. The proposed hybrid framework successfully identified these characteristics using transformer-based semantic analysis and NLP feature extraction techniques.

The system also demonstrated robustness against paraphrased and prompt-engineered AI-generated content.

4.3 System Behavior

The framework successfully demonstrated contextual semantic analysis and AI-generated text classification across multiple datasets and writing styles.

- NLP preprocessing improved textual consistency.
- TF-IDF vectorization captured statistical features.
- Transformer embeddings improved semantic understanding.
- Topic-based splitting improved robustness and generalization

CHAPTER 5

CONCLUSION AND FUTURE WORK

The project “AI-Generated Text Detection Using Hybrid NLP and Transformer Models” successfully demonstrates a scalable and robust framework for identifying AI-generated content using hybrid NLP techniques and transformer-based semantic analysis.

The integration of stylometric analysis, TF-IDF vectorization, and transformer embeddings significantly improved contextual understanding, robustness, and classification accuracy across multiple benchmark datasets. Transformer-based architectures such as DistilBERT and RoBERTa outperformed traditional machine learning approaches in semantic understanding and generalization capability.

The proposed framework provides an effective solution for academic integrity verification, misinformation detection, plagiarism prevention, and digital content authentication systems.

Future Work

- Real-time AI-generated content monitoring systems.
- Multilingual AI-generated text detection.
- Lightweight transformer optimization.
- Explainable AI-based detection dashboards.
- Adversarial robustness improvement.
- Integration with educational verification platforms.

REFERENCES

- 1.Devlin et al., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” NAACL-HLT, 2019.
- 2.Sanh et al., “DistilBERT: Smaller, Faster, Cheaper and Lighter,” 2019.
- 3.Liu et al., “RoBERTa: A Robustly Optimized BERT Pretraining Approach,” 2019.
- 4.OpenAI, “GPT-4 Technical Report,” 2023.
- Mitchell et al., “DetectGPT: Zero-Shot Machine-Generated Text Detection,” ICML, 2023.
- 5.Weber-Wulff et al., “Testing of Detection Tools for AI-Generated Text,” 2023.
- 6.Zhang and Zhao, “Hybrid NLP and Transformer Models for AI Text Detection,” IEEE Access, 2023.
- 7.Gehrmann et al., “GLTR: Statistical Detection and Visualization of Generated Text,” ACL, 2019.